12598. *Proposed by Paul Bracken, University of Texas, Edinburg, TX.*

(a) Evaluate

$$\sum_{n=1}^{\infty}(-1)^{n-1}\left(\sum_{k=0}^{\infty}\frac{(-1)^k}{n+2k}\right).$$

(b) Evaluate

$$\sum_{n=1}^{\infty}(-1)^{n-1}\left(\sum_{k=0}^{\infty}\frac{(-1)^k}{n+2k}\right)^2.$$

Solution to Problem 12598 in Amer. Math. Monthly 133 (2026)

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We show that the desired sums are $\ln(2)/4+\pi/8$ and $\pi^2/32+\pi/8-\ln(2)/4$, respectively.

(a) Let

$$S := \sum_{n=1}^{\infty}(-1)^{n-1}\left(\sum_{k=0}^{\infty}\frac{(-1)^k}{n+2k}\right).$$

Then, we can rewrite S to get

$$S = \sum_{n=1}^{\infty}\sum_{k=0}^{\infty}\frac{(-1)^{n-1}(-1)^k}{n+2k} = \sum_{n=1}^{\infty}\sum_{k=0}^{\infty}\int_0^{\infty}(-1)^{n-1}(-1)^k e^{-(n+2k)t}dt.$$

To justify the interchange of integration and double summation, we apply the dominated convergence theorem twice. For the first swap, we define the sequence of functions

$$f_{n,K}(t) := \sum_{k=0}^{K}(-1)^{n-1}(-1)^k e^{-(n+2k)t}dt.$$

We note that as $K \to \infty$ the sequence of functions converges pointwise for all $t > 0$.

Then

$$\begin{aligned}|f_{n,K}(t)| &= \left|\sum_{k=0}^{K}(-1)^{n-1}(-1)^k e^{-(n+2k)t}\right| = \left|e^{-nt}\cdot\frac{1-(-e^{-2t})^{(K+1)}}{1+e^{-2t}}\right| \\ &\le \left|e^{-nt}\cdot\frac{1+e^{-2t(K+1)}}{1+e^{-2t}}\right| = e^{-nt}\cdot\frac{1+e^{-2t(K+1)}}{1+e^{-2t}} \le e^{-nt}\end{aligned}$$

for all $n \geq 1$, $K \geq 0$, and $t \geq 0$. Additionally, $\int_0^\infty e^{-nt}dt = 1/n$ for all $n \geq 1$. Thus, by the dominated convergence theorem,

$$S = \sum_{n=1}^{\infty} \int_0^\infty \sum_{k=0}^{\infty} (-1)^{n-1}(-1)^k e^{-(n+2k)t} dt.$$

To justify the second swap, we define a sequence of functions

$$m_N(t) := \sum_{n=1}^{N} \sum_{k=0}^{\infty} (-1)^{n-1}(-1)^k e^{-(n+2k)t} dt.$$

We note that as $N \to \infty$ the sequence converges pointwise for all $t > 0$.

Then

$$\begin{aligned} |m_N(t)| &= \left| \sum_{n=1}^{N} \sum_{k=0}^{\infty} (-1)^{n-1}(-1)^k e^{-(n+2k)t} \right| = \left| e^{-t} \cdot \frac{1 - (-e^{-t})^N}{(1+e^{-t})(1+e^{-2t})} \right| \\ &\leq \left| e^{-t} \cdot \frac{1 + e^{-Nt}}{(1+e^{-t})(1+e^{-2t})} \right| = e^{-t} \cdot \frac{1 + e^{-Nt}}{(1+e^{-t})(1+e^{-2t})} \leq e^{-t} \end{aligned}$$

for all $N \geq 1$ and $t \geq 0$. Additionally, $\int_0^\infty e^{-t}dt = 1$. Thus, by the dominated convergence theorem,

$$S = \int_0^\infty \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} (-1)^{n-1}(-1)^k e^{-(n+2k)t} dt.$$

Evaluating the double sum, we get

$$\begin{aligned} S &= \int_0^\infty \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} (-1)^{n-1}(-1)^k e^{-(n+2k)t} dt \\ &= \int_0^\infty \left(\sum_{n=1}^{\infty} (-1)^{n-1} e^{-nt} \right) \left(\sum_{k=0}^{\infty} (-1)^k e^{-2kt} \right) dt \\ &= \int_0^\infty \left(\frac{e^{-t}}{1+e^{-t}} \right) \left(\frac{1}{1+e^{-2t}} \right) dt \\ &= \int_0^\infty \frac{e^{-t}}{(1+e^{-t})(1+e^{-2t})} dt. \end{aligned}$$

Now, we rearrange the integrand and perform the substitution $x = 1 + e^{-t}$ to obtain

$$\int_0^\infty \frac{e^{-t}}{(1+e^{-t})(1+e^{-2t})}dt = -\int_0^\infty \frac{-e^{-t}}{(1+e^{-t})((1+e^{-t})(-1+e^{-t})+2)}dt$$
$$= -\int_2^1 \frac{1}{x(x(x-2)+2)}dx$$
$$= \int_1^2 \frac{1}{x(x^2-2x+2)}dx.$$

We then decompose the integrand to obtain

$$\int_1^2 \frac{1}{x(x^2-2x+2)}dx = \int_1^2 \left(\frac{1}{2x} + \frac{1}{x^2-2x+2} - \frac{x}{2(x^2-2x+2)}\right)dx$$
$$= \frac{1}{2}\int_1^2 \frac{1}{x}dx + \int_1^2 \frac{1}{x^2-2x+2}dx - \frac{1}{2}\int_1^2 \frac{x}{x^2-2x+2}dx.$$

We denote the result as $I_1 + I_2 + I_3$.

Evaluating I_1, we get

$$I_1 = \frac{1}{2}\int_1^2 \frac{1}{x}dx = \frac{1}{2}(\ln(2) - \ln(1)) = \frac{\ln(2)}{2}.$$

Evaluating I_2, we get

$$I_2 = \int_1^2 \frac{1}{x^2-2x+2}dx = \int_1^2 \frac{1}{(x-1)^2+1}dx$$
$$= \arctan(2-1) - \arctan(1-1) = \arctan(1) = \frac{\pi}{4}.$$

To evaluate I_3, we begin by integrating by parts with $u = x$ and $v' = 1/(x^2-2x+2)$. Hence,

$$I_3 = -\frac{1}{2}\int_1^2 \frac{x}{x^2-2x+2}dx = -\frac{x\arctan(x-1)}{2}\Big|_1^2 + \frac{1}{2}\int_1^2 \arctan(x-1)dx$$
$$= -\frac{\pi}{4} + \frac{1}{2}\int_1^2 \arctan(x-1)dx.$$

For the remaining integral, we make the substitution $x = \tan(\theta) + 1$ and obtain

$$\frac{1}{2}\int_1^2 \arctan(x-1)dx = \frac{1}{2}\int_0^{\frac{\pi}{4}} \arctan(\tan(\theta))\sec^2(\theta)d\theta = \frac{1}{2}\int_0^{\frac{\pi}{4}} \theta\sec^2(\theta)d\theta.$$

Then, we integrate by parts with $u = \theta$ and $v' = \sec^2(\theta)$ to get

$$\begin{aligned}
\frac{1}{2}\int_0^{\frac{\pi}{4}} \theta \sec^2(\theta) d\theta &= \frac{1}{2}\left(\theta \tan\theta\Big|_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan\theta d\theta\right) \\
&= \frac{1}{2}\left(\theta \tan\theta\Big|_0^{\frac{\pi}{4}} + \ln(|\cos\theta|)\Big|_0^{\frac{\pi}{4}}\right) \\
&= \frac{1}{2}\left(\frac{\pi}{4} - \frac{\ln(2)}{2}\right) \\
&= \frac{\pi}{8} - \frac{\ln(2)}{4}.
\end{aligned}$$

Thus, $I_3 = -\pi/4 + \pi/8 - \ln(2)/4$ and

$$I_1 + I_2 + I_3 = \frac{\ln(2)}{2} + \frac{\pi}{4} + \frac{\pi}{8} - \frac{\ln(2)}{4} - \frac{\pi}{4} = \frac{\ln(2)}{4} + \frac{\pi}{8}.$$

Therefore,

$$S = \sum_{n=1}^{\infty} (-1)^{n-1} \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{n+2k}\right) = \frac{\ln(2)}{4} + \frac{\pi}{8}.$$

(b) Let

$$S := \sum_{n=1}^{\infty} (-1)^{n-1} \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{n+2k}\right)^2$$

and

$$J_n := \sum_{k=0}^{\infty} \frac{(-1)^k}{n+2k}.$$

Then

$$S = \sum_{n=1}^{\infty} (-1)^{n-1} J_n^2.$$

Additionally, J_n can be rewritten as

$$\sum_{k=0}^{\infty} \int_0^{\infty} (-1)^k e^{-(n+2k)s} ds.$$

To justify swapping integration and summation, we take a similar approach as used to justify the first swap in part **(a)** and define a sequence of functions

$$f_{n,K}(s) := \sum_{k=0}^{K}(-1)^k e^{-(n+2k)s} ds.$$

We note that as $K \to \infty$ the sequence converges pointwise for all $s > 0$.

Then

$$\begin{aligned} |f_{n,K}(s)| &= \left|\sum_{k=0}^{K}(-1)^k e^{-(n+2k)s}\right| = \left|e^{-ns} \cdot \frac{1-(-e^{-2s})^{(K+1)}}{1+e^{-2s}}\right| \\ &\leq e^{-ns} \cdot \left|\frac{1+e^{-2s(K+1)}}{1+e^{-2s}}\right| \leq e^{-ns} \end{aligned}$$

for all $K \geq 0$ and $s \geq 0$. Moreover, $\int_0^\infty e^{-ns} ds = 1/n$. Thus, by the dominated convergence theorem,

$$\sum_{k=0}^{\infty}\int_0^\infty (-1)^k e^{-(n+2k)s} ds = \int_0^\infty \sum_{k=0}^{\infty}(-1)^k e^{-(n+2k)s} ds.$$

Hence,

$$\begin{aligned} J_n^2 = J_n \cdot J_n &= \left(\int_0^\infty \sum_{k=0}^{\infty}(-1)^k e^{-(n+2k)s} ds\right)\left(\int_0^\infty \sum_{k=0}^{\infty}(-1)^k e^{-(n+2k)t} dt\right) \\ &= \int_0^\infty \int_0^\infty \left(\sum_{k=0}^{\infty}(-1)^k e^{-(n+2k)s}\right)\left(\sum_{k=0}^{\infty}(-1)^k e^{-(n+2k)t}\right) dsdt \\ &= \int_0^\infty \int_0^\infty \left(\frac{e^{-ns}}{1+e^{-2s}}\right)\left(\frac{e^{-nt}}{1+e^{-2t}}\right) dsdt \\ &= \int_0^\infty \int_0^\infty \frac{e^{-n(s+t)}}{(1+e^{-2s})(1+e^{-2t})} dsdt. \end{aligned}$$

Substituting J_n^2 into S, we get

$$\begin{aligned} S &= \sum_{n=1}^{\infty}(-1)^{n-1}\int_0^\infty \int_0^\infty \frac{e^{-n(s+t)}}{(1+e^{-2s})(1+e^{-2t})} dsdt \\ &= \sum_{n=1}^{\infty}\int_0^\infty \int_0^\infty \frac{(-1)^{n-1}e^{-n(s+t)}}{(1+e^{-2s})(1+e^{-2t})} dsdt. \end{aligned}$$

To justify swapping summation and double integration, we use the dominated convergence theorem again.

We define a sequence of multivariate functions

$$f_N(s,t) = \sum_{n=1}^{N} \frac{(-1)^{n-1} e^{-n(s+t)}}{(1+e^{-2s})(1+e^{-2t})}.$$

.

We note that as $N \to \infty$ the sequence converges pointwise for all $s, t > 0$.

Then,

$$\begin{aligned}
\left| \sum_{n=1}^{N} \frac{(-1)^{n-1} e^{-n(s+t)}}{(1+e^{-2s})(1+e^{-2t})} \right| &\leq \left| e^{-(s+t)} \cdot \frac{1 - \left(-e^{-(s+t)}\right)^N}{(1+e^{-(s+t)})(1+e^{-2s})(1+e^{-2t})} \right| \\
&\leq \left| e^{-(s+t)} \cdot \frac{1 + e^{-N(s+t)}}{(1+e^{-(s+t)})(1+e^{-2s})(1+e^{-2t})} \right| \\
&= e^{-(s+t)} \cdot \frac{1 + e^{-N(s+t)}}{(1+e^{-(s+t)})(1+e^{-2s})(1+e^{-2t})} \\
&\leq e^{-(s+t)}
\end{aligned}$$

for all $N \geq 1$ and $s, t \geq 0$. Additionally, $\int_0^\infty \int_0^\infty e^{-(s+t)} ds dt = 1$. Thus, by the dominated convergence theorem,

$$S = \int_0^\infty \int_0^\infty \sum_{n=1}^{\infty} \frac{(-1)^{n-1} e^{-n(s+t)}}{(1+e^{-2s})(1+e^{-2t})} ds dt.$$

Evaluating the sum, we obtain

$$\begin{aligned}
S &= \int_0^\infty \int_0^\infty \sum_{n=1}^{\infty} \frac{(-1)^{n-1} e^{-n(s+t)}}{(1+e^{-2s})(1+e^{-2t})} ds dt \\
&= \int_0^\infty \int_0^\infty \frac{e^{-(s+t)}}{(1+e^{-(s+t)})(1+e^{-2s})(1+e^{-2t})} ds dt.
\end{aligned}$$

Now, we perform the transformation $s = -\ln u$, $t = -\ln v$. The Jacobian determinant $|\partial(s,t)/\partial(u,v)|$ of this transformation is

$$\begin{vmatrix} -\frac{1}{u} & 0 \\ 0 & -\frac{1}{v} \end{vmatrix} = \frac{1}{uv}.$$

Hence,

$$
\begin{aligned}
S &= \int_0^\infty \int_0^\infty \frac{e^{-(s+t)}}{(1+e^{-(s+t)})(1+e^{-2s})(1+e^{-2t})} dsdt \\
&= \int_1^0 \int_1^0 \frac{uv}{(1+uv)(1+u^2)(1+v^2)} \cdot \frac{1}{uv} dudv \\
&= \int_0^1 \int_0^1 \frac{1}{(1+uv)(1+u^2)(1+v^2)} dudv.
\end{aligned}
$$

Decomposing the integrand, we get

$$
\begin{aligned}
S &= \int_0^1 \int_0^1 \left(\frac{v^2}{(1+uv)(1+v^2)^2} + \frac{1}{(1+u^2)(1+v^2)^2} - \frac{uv}{(1+u^2)(1+v^2)^2} \right) dudv \\
&= \int_0^1 \int_0^1 \frac{1}{(1+u^2)(1+v^2)^2} dudv - \int_0^1 \int_0^1 \frac{uv}{(1+u^2)(1+v^2)^2} dudv \\
&\quad + \int_0^1 \int_0^1 \frac{v^2}{(1+uv)(1+v^2)^2} dudv.
\end{aligned}
$$

We denote the result as $I_1 + -I_2 + I_3$.

Simplifying I_1, we obtain

$$
\begin{aligned}
I_1 &= \int_0^1 \int_0^1 \frac{1}{(1+u^2)(1+v^2)^2} dudv \\
&= \left(\int_0^1 \frac{1}{1+u^2} du \right) \left(\int_0^1 \frac{1}{(1+v^2)^2} dv \right) \\
&= \arctan(1) \cdot \int_0^1 \frac{1}{(1+v^2)^2} dv \\
&= \frac{\pi}{4} \int_0^1 \frac{1}{(1+v^2)^2} dv.
\end{aligned}
$$

Now, we perform the substitution $v = \tan\theta$ and evaluate the integral to get

$$\begin{aligned}
\frac{\pi}{4}\int_0^1 \frac{1}{(1+v^2)^2}dv &= \frac{\pi}{4}\int_0^{\frac{\pi}{4}} \frac{\sec^2\theta}{(1+\tan^2\theta)^2}d\theta \\
&= \frac{\pi}{4}\int_0^{\frac{\pi}{4}} \frac{\sec^2\theta}{(\sec^2\theta)^2}d\theta \\
&= \frac{\pi}{4}\int_0^{\frac{\pi}{4}} \frac{1}{\sec^2\theta}d\theta \\
&= \frac{\pi}{4}\int_0^{\frac{\pi}{4}} \frac{1+\cos 2\theta}{2}d\theta \\
&= \frac{\pi}{4}\Big(\frac{2\theta+\sin 2\theta}{4}\Big|_0^{\frac{\pi}{4}}\Big) \\
&= \frac{\pi}{4}\left(\frac{\pi}{8}+\frac{1}{4}\right) = \frac{\pi^2}{32}+\frac{\pi}{16}.
\end{aligned}$$

Thus, $I_1 = \pi^2/32 + \pi/16$.

Evaluating $-I_2$, we obtain

$$\begin{aligned}
-\int_0^1\int_0^1 \frac{uv}{(1+u^2)(1+v^2)^2}dudv &= -\left(\int_0^1 \frac{u}{1+u^2}du\right)\left(\int_0^1 \frac{v}{(1+v^2)^2}dv\right) \\
&= -\Big(\frac{1}{2}\ln(1+u^2)\Big|_0^1\Big)\cdot\Big(-\frac{1}{2(1+v^2)}\Big|_0^1\Big) \\
&= -\frac{\ln(2)}{2}\cdot\frac{1}{4} = -\frac{\ln(2)}{8}.
\end{aligned}$$

To evaluate I_3, we perform the transformation $u = (x-1)/y$, $v = y$. The Jacobian determinant $|\partial(u,v)/\partial(x,y)|$ of the transformation is

$$\begin{vmatrix} \frac{1}{y} & -\frac{(x-1)}{y^2} \\ 0 & 1 \end{vmatrix} = \frac{1}{y}.$$

Therefore,

$$\begin{aligned}
I_3 = \int_0^1\int_0^1 \frac{v^2}{(1+uv)(1+v^2)^2}dudv &= \int_0^1\int_1^{y+1} \frac{y}{x(1+y^2)^2}dxdy \\
&= \int_0^1 \frac{y}{(1+y^2)^2}\left(\int_1^{y+1}\frac{1}{x}dx\right)dy.
\end{aligned}$$

Evaluating the inner integral, we obtain

$$\int_0^1 \frac{y}{1+y^2}\left(\int_1^{y+1}\frac{1}{x}dx\right)dy = \int_0^1 \frac{y\ln(y+1)}{(1+y^2)^2}dy.$$

Then, we integrate by parts with $u = \ln(y+1)$ and $v' = y/(1+y^2)^2$ to obtain

$$\begin{aligned}\int_0^1 \frac{y\ln(y+1)}{(1+y^2)^2}dy &= -\frac{\ln(y+1)}{2(1+y^2)}\Big|_0^1 + \frac{1}{2}\int_0^1 \frac{1}{(1+y)(1+y^2)}dy \\ &= -\frac{\ln(2)}{4} + \frac{1}{2}\int_0^1 \frac{1}{(1+y)(1+y^2)}dy.\end{aligned}$$

We then partially decompose the integrand of the remaining integral to obtain

$$\begin{aligned}\frac{1}{2}\int_0^1 \frac{1}{(1+y)(1+y^2)}dy &= \frac{1}{2}\int_0^1 \left(\frac{1}{2(1+y)} - \frac{y}{2(1+y^2)} + \frac{1}{2(1+y^2)}\right)dy \\ &= \frac{1}{4}\int_0^1 \frac{1}{1+y}dy - \frac{1}{4}\int_0^1 \frac{y}{1+y^2}dy + \frac{1}{4}\int_0^1 \frac{1}{1+y^2}dy \\ &= \frac{1}{4}\Big(\ln(1+y)\Big|_0^1\Big) - \frac{1}{4}\left(\frac{\ln(1+y^2)}{2}\Big|_0^1\right) + \frac{1}{4}\Big(\arctan(y)\Big|_0^1\Big) \\ &= \frac{\ln(2)}{4} - \frac{\ln(2)}{8} + \frac{\pi}{16} \\ &= \frac{\ln(2)}{8} + \frac{\pi}{16}.\end{aligned}$$

Thus, $I_3 = -\ln(2)/4 + \ln(2)/8 + \pi/16$ and

$$I_1 + -I_2 + I_3 = \frac{\pi^2}{32} + \frac{\pi}{16} - \frac{\ln(2)}{8} - \frac{\ln(2)}{4} + \frac{\ln(2)}{8} + \frac{\pi}{16} = \frac{\pi^2}{32} + \frac{\pi}{8} - \frac{\ln(2)}{4}.$$

Therefore,

$$S = \sum_{n=1}^{\infty}(-1)^{n-1}\left(\sum_{k=0}^{\infty}\frac{(-1)^k}{n+2k}\right)^2 = \frac{\pi^2}{32} + \frac{\pi}{8} - \frac{\ln(2)}{4}.$$